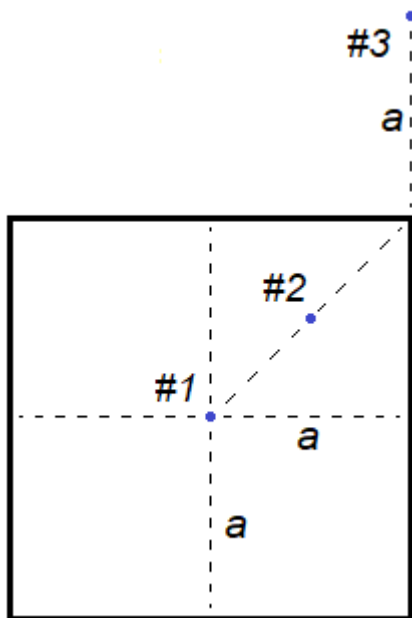


The objective of this lab exercise is to measure and analyze the magnetic field created by a square loop of current.

Using the Biot-Savart Law, we can derive an expression for the magnetic field created by a square loop of current at three points, as shown:



Point #1 is at the center of the square. Point #2 is midway between the center and one corner of the square. Point #3 is outside of the square, at a distance a from one corner.

The magnitude of the magnetic field at each of these points can be expressed as:

$$B = \alpha \frac{Nk_m}{2a} I$$

where N is the number of turns and α is a unitless number that depends on the location of the point. For the three points shown, these values are:

Point #1 $\alpha = 5.66$

Point #2 $\alpha = 7.99$

Point #3 $\alpha = 0.52$

Since k , α , N , and a are constants, we should find that B is proportional to I , so that a graph of B vs. I should result in a straight line with a slope of:

$$\alpha \frac{Nk_m}{2a}$$

Procedure

Record the value of N . Measure and record the value of a .

On the paper taped to the table, trace the position of the square coil and carefully mark the locations of Points 1 2 and 3.

For ten values of current – 0.2, 0.4, 0.6, ... , 2.0 Amps – measure the magnetic field at each of the three locations

Plot all three sets of data, i.e. B vs I , on one graph; find the equation of the best fit straight line to each set of data.

Calculate the expected value of the slope for each of the three sets of data, and compare the measured value of each set to the corresponding expected value.